
Mode Collapse under Dr.GRPO: MODEBENCH and Online Verified Maximum Entropy

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Abstract

Binary verifier rewards tell a reasoning policy which responses are correct, but not which correct responses are redundant. Under Dr.GRPO, this missing signal can produce *mode collapse*: probability concentrates on a strict subset of correct outcomes even while accuracy is stable or improving. Token entropy does not diagnose the failure because many strings can encode the same solution. We formalize collapse on the conditional distribution of verifier-positive execution modes and introduce MODEBENCH, four environments in which the same executable validator establishes both correctness and mode identity. We then propose xDr.GRPO, an online verified MaxEnt-Dr.GRPO extension. It builds support only from policy-generated, validator-positive outcomes; rewards rare and first-seen modes with an open-set advantage; and globally replays discovered verified exemplars under adaptive mass and balance terms. At the largest preregistered checkpoint shared by all three seeds of both arms, xDr.GRPO improves pass@8 over matched Dr.GRPO in all four domains: .372 $\rightarrow$.968, .608 $\rightarrow$.665, .172 $\rightarrow$.740, and .364 $\rightarrow$.779. More importantly, additional correct modes beyond the first rise from .021, .042, .000, .000 to 1.376, 1.010, .216, .026. This broad comparison is exploratory because the registered ModeBench campaign was not completed. A separate terminal MathIR ablation shows that isolated support discovery raises distinct@8 by .137 and pass@8 by .027. Together, these results support an executable benchmark, an online outcome-entropy intervention, and causal evidence for separated discovery.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) can make a reasoning model more accurate while making repeated samples less useful. A binary verifier rewards every correct response equally. Dr.GRPO removes a length bias from GRPO, but it does not add a force that distinguishes two correct solutions [10]. If one correct solution is sampled more often, it receives more updates; if another disappears from the on-policy group, it receives none. Eventually, eight draws can become eight renderings of one outcome. We call this failure *mode collapse under Dr.GRPO*.

The term needs a precise object. Surface strings are not modes: changes in spacing, derivation length, or prose can leave the executed solution unchanged. Final answers are not sufficient either: two algorithms can return the same answer, while two different-looking expressions can be algebraic aliases. For prompt x , we therefore define a mode by an executable validator $V(y, s_x)$. It returns $\perp$ for an invalid response and a canonical key c for a correct executed outcome. Mode collapse is concentration of the conditional distribution over these validator-positive keys, not a decrease in token entropy and not necessarily a decrease in correctness.

This distinction matters for test-time scaling. Parallel sampling, self-consistency, search, and self-improvement benefit only when extra samples cover alternatives rather than repeat the same solution [14, 13, 8]. It also changes how collapse should be measured. Let $D_K(x)$ be the number of distinct

correct keys in K samples and $P_K(x) = \mathbf{1}[D_K(x) > 0]$. Then

$$\text{excess@}K := \mathbb{E}[D_K - P_K] = \text{distinct@}K - \text{pass@}K.$$

A policy that produces at most one correct mode has $\text{excess@}K = 0$ regardless of its accuracy. This simple diagnostic is the organizing metric of the paper. Figure 1 grounds it in matched model trajectories: Dr.GRPO’s correct samples rise to 32 out of 32 precisely while its sampled correct support contracts to one repeated coloring; under the same checkpoints, xDr.GRPO retains multimodal support.

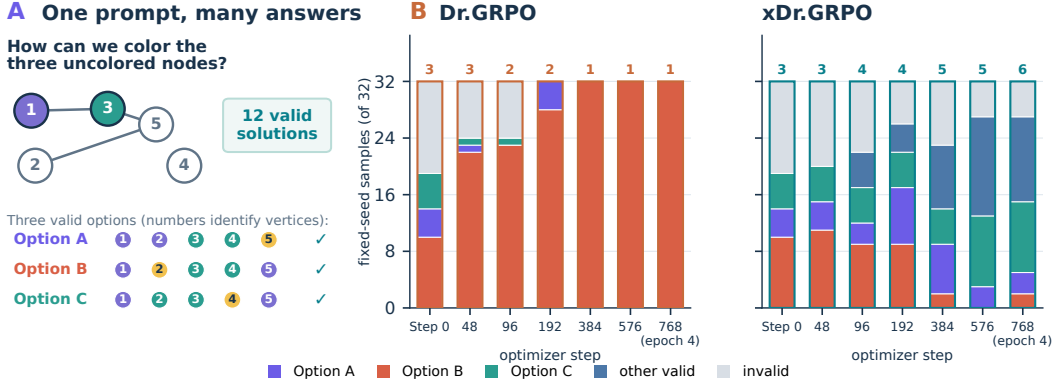


Figure 1: **Mode-collapse example.** (A) The prompt has 12 valid solutions; three are shown as Options A–C. Numbers inside circles identify vertices, while circle fill denotes assigned color. (B) Separate Dr.GRPO and xDr.GRPO facets show matched Qwen2.5-0.5B-Instruct trajectories (seed 43), never paired bars on one axis. Each bar pools four fixed-seed, eight-sample draws: colored segments are correct modes, gray is invalid output, and the number above each bar is the observed correct-mode count. The x-axis reports optimizer steps and stops at step 768, the end of epoch 4; no epoch-5 checkpoint is shown. Dr.GRPO reaches 32/32 correct but collapses to one mode by step 384 and remains there, while xDr.GRPO retains six modes at step 768. Aggregate results are in Table 3 and Figure 2.

We make the problem experimentally observable with MODEBENCH. Its four domains span a constraint assignment, an arithmetic expression, a program behavior, and an algebraic procedure. In each, correctness and identity come from the same executed object. Formatting aliases merge, different valid executions remain distinct, and training never receives a gold catalogue of modes.

We address the failure with xDr.GRPO, short for *executable-mode Dr.GRPO*: an online verified outcome-MaxEnt extension of Dr.GRPO.¹ The learner records only correct modes that its current policy actually generates. An open-set predictive advantage rewards rare modes, a one-time credit rewards first discovery, and a global scheduler makes one verified bank act on every optimizer update through verified-mass and known-mode-balance replay. The task reward and Dr.GRPO normalization remain unchanged.

Our claims follow the evidence rather than the size of the experiment archive. The four-domain method comparison is a frozen, three-seed shared-checkpoint analysis and is explicitly exploratory. A six-run terminal MathIR ablation supports a narrower causal claim about separated support discovery.

Our contributions are:

- an operational definition of mode collapse under Dr.GRPO, including a diagnostic that separates correctness from breadth;
- MODEBENCH, a four-domain executable benchmark in which reward and semantic identity are bound to one validation path;

¹The “x” denotes executable outcomes. It does not denote the detached candidate-reweighting surrogate called xDr in preliminary repository experiments; those runs are not pooled into any result in this paper.

- xDr.GRPO, an online verified MaxEnt-Dr.GRPO method over policy-discovered correct outcomes, without a gold support catalogue; and
- an auditable evidence hierarchy: a broad exploratory comparison, a positive terminal causal ablation, and explicit exclusion of incomplete or failed cohorts.

2 Mode Collapse under Dr.GRPO

2.1 Correct mass and correct-mode distribution

For prompt x , response y , and reference specification s_x , let

$$V(y, s_x) \in \{\perp\} \cup \mathcal{C}_x, \quad r(x, y) = \mathbf{1}[V(y, s_x) \neq \perp].$$

The policy’s total correct mass is $P_\theta^+(x) = \Pr_{Y \sim \pi_\theta}[V(Y, s_x) \neq \perp]$. Conditional on correctness, its mode distribution is

$$p_\theta^+(c \mid x) = \Pr[V(Y, s_x) = c \mid V(Y, s_x) \neq \perp].$$

We say training exhibits *mode collapse* on x when probability within p_θ^+ concentrates on a strict subset of $\mathcal{C}_x$ while P_θ^+ remains comparable or increases. Low accuracy alone is not mode collapse; a high-accuracy policy supported on one correct mode is.

Two population quantities describe the failure: conditional verified-mode entropy $H_\theta^+(x) = -\sum_c p_\theta^+(c \mid x) \log p_\theta^+(c \mid x)$ and concentration $\max_c p_\theta^+(c \mid x)$. They are unavailable when the support is unknown, so evaluation uses samples:

$$\begin{aligned} \text{pass@}K &= \mathbb{E}_x \mathbb{E}[P_K(x)], \\ \text{distinct@}K &= \mathbb{E}_x \mathbb{E}[D_K(x)], \\ \text{excess@}K &= \mathbb{E}_x \mathbb{E}[D_K(x) - P_K(x)]. \end{aligned}$$

The first measures whether search finds any correct mode; the second counts correct modes; the third removes the first success and isolates breadth beyond it. We also report $\text{mean@}K$, the fraction of the K draws that are correct, and greedy $\text{pass@}1$. No one metric substitutes for the others.

2.2 Why equal reward does not preserve equal modes

For a group of G responses, Dr.GRPO uses the task advantage

$$A_i^{\text{task}} = r_i - \frac{1}{G} \sum_{j=1}^G r_j$$

with one shared maximum-length normalization. Correct responses in the same group receive the same task advantage. That symmetry removes a preference based on identity, but it does not preserve mode ratios.

A categorical toy model makes the feedback explicit. Let z_c be a logit for correct mode c , $p_c = \text{softmax}(z)_c$, and $P^+ = \sum_{c \in \mathcal{C}} p_c$. Gradient ascent on expected binary reward gives

$$\dot{z}_c = \frac{\partial P^+}{\partial z_c} = p_c(1 - P^+), \quad \frac{d}{dt} \log \frac{p_c}{p_d} = (p_c - p_d)(1 - P^+).$$

While incorrect mass remains, the more common of two equally rewarded correct modes receives the larger logit update. Finite on-policy groups sharpen the asymmetry: an absent rare mode receives no update, and an all-correct group has zero centered task advantage. Thus the first successful mode can become a self-reinforcing and eventually locally frozen solution. This idealized calculation is a mechanism, not a claim that a language model is a categorical bandit; it explains why binary reward alone supplies no restoring force. Recent analyses reach related conclusions through outcome-frequency amplification, divergence choice, or set-level policy geometry [12, 9, 7].

Appendix A makes the mechanism precise. Under the infinitesimal on-policy mean dynamics, Theorem A.2 shows that generic binary-reward GRPO and Dr.GRPO trajectories select one correct mode. After the verified bank covers the correct support, Theorem A.5 and Corollary A.6 show that xDr.GRPO reverses the unstable mode-ratio dynamics and rules out collapse in the categorical abstraction. The coverage and optimization assumptions are stated explicitly; the theorem does not pretend that an arbitrary neural policy must discover an unseen mode.

3 MODEBENCH: Executable Mode Measurement

3.1 Benchmark contract

A benchmark instance is mode-verifiable only when V satisfies two requirements. First, it validates the object the model actually generated, not a model-supplied claim about that object. Second, the canonical key is derived from the same execution that established correctness. Cosmetic aliases must merge; different executed outcomes must remain distinct. This contract prevents wrong-but-novel responses from receiving diversity credit and prevents formatting noise from masquerading as solution breadth.

Domain	Generated object / check	Mode key	Why it is semantic	train/eval
Graph coloring	Complete assignment; check fixed colors and every edge	Color vector	Spacing aliases merge; valid assignments differ	192/96
Countdown	Parse AST; enforce exact operands; execute rationally	Normalized AST	Commutative aliases merge; expression trees differ	384/128
Python factors	Call restricted function in an external worker	Returned integer vector	Source aliases merge; behaviors differ	384/128
MathIR	Execute a bounded algebra action program exactly	Equation-state trajectory	Endpoints can match while routes differ	384/128

Table 1: **The MODEBENCH contract.** Correctness and mode identity are outputs of the same executable validation path. No exhaustive mode catalogue initializes training.

Graph coloring. Each prompt specifies a graph, a partial three-color assignment, and hidden vertices. The validator reconstructs the complete assignment, preserves shown colors, and rejects every monochromatic edge. The full color vector is the key.

Countdown. Each prompt gives three distinct integers and a target. A response must use each integer exactly once with $+$, $-$, $\times$, $\div$ and parentheses. The parser executes with exact rational arithmetic. Canonicalization sorts commutative children, merging syntactic aliases but retaining genuinely different trees.

Python factors. Each prompt asks for `lambda n: EXPR` on four composite inputs. A syntax-restricted external Python worker requires every returned d to satisfy $1 < d < n$ and $n \bmod d = 0$. The complete output vector is the key, so different program texts with identical behavior merge. A prompt with input tuple $(n_1, \dots, n_m)$ has $\prod_j |D(n_j)|$ exact modes, where $D(n)$ is the set of proper nontrivial divisors. The released rows contain 16–3,600 modes each.

MathIR. Each prompt shows a linear equation and at most eight lettered transformations. The model emits at most four actions. An interpreter applies each action to both sides and normalizes with exact rational arithmetic after every step. The key is the resulting state trajectory. Each prompt has five exhaustively enumerated valid routes, withheld from the training policy and bank.

4 xDr.GRPO: Online Verified MaxEnt-Dr.GRPO

xDr.GRPO is an outcome-level maximum-entropy surrogate layered on Dr.GRPO. It does not maximize token entropy and does not assume the valid support is known. It couples three signals: online rare/new-mode advantages, verified replay, and self-referenced adaptive coefficients. The task advantage remains unchanged.

4.1 Online verified support

For each prompt x , the learner maintains a bank $\mathcal{B}_x^+$ of keys that the current policy generated and V accepted. It starts empty. Every group is scored against one immutable pre-group snapshot, and new keys are committed after the whole group, making novelty invariant to row order. Admission is fail-closed at the intersection of positive rollout reward and executable canonical validation. A disagreement receives no key and is recorded.

The matched Dr.GRPO control runs the same validator and passive counts with all auxiliary coefficients set to zero. Thus discovery telemetry is comparable without changing its optimization objective.

4.2 Open-set rare-mode and discovery advantages

For a verified mode a , let $c_x(a)$ combine its historical count with its leave-one-out multiplicity among the other verified responses in the group. With pseudocount $\lambda = 1$, the detached predictive distribution includes one structural unseen bucket $\emptyset$:

$$q_x(a) = \frac{c_x(a) + \lambda}{\sum_b c_x(b) + \lambda(K_x + 1)}, \quad q_x(\emptyset) = \frac{\lambda}{\sum_b c_x(b) + \lambda(K_x + 1)}.$$

Here K_x is the number of explicitly counted keys. The unseen bucket says the observed bank is not assumed complete.

With surprisal clip $S = 5$, define

$$h_i = \min\{-\log q_x(a_i), S\}, \quad \bar{h}_q = \mathbb{E}_{a \sim q_x} \min\{-\log q_x(a), S\},$$

$$A_i^{\text{rare}} = \beta_t \frac{h_i - \bar{h}_q}{S}.$$

Centering under q_x , rather than under the empirical group, keeps a carrier when the current correct samples all share one mode but the bank records more.

Let $\mathcal{N}_x$ be the verified keys absent from the pre-group bank and $m_x(a)$ their current-group multiplicity. A first discovery receives

$$A_i^{\text{new}} = \nu \frac{\mathbf{1}[a_i \in \mathcal{N}_x]}{m_x(a_i)}, \quad \nu = 0.50.$$

All duplicates of one new key share total credit ν . The actor advantage is

$$A_i = A_i^{\text{task}} + \mathbf{1}[V(y_i, s_x) \neq \perp] (A_i^{\text{rare}} + A_i^{\text{new}}).$$

There is no second empirical centering after the mode terms. Invalid responses receive neither novelty nor entropy pressure.

4.3 Global verified replay

With hundreds of training prompts, a prompt-local discovery may otherwise wait nearly a pass before acting again. After the first verified bank exists, every optimizer update therefore schedules exactly one bank in deterministic global round-robin order. It retains at most one policy-generated exemplar per observed key, up to 16 keys. Two losses apply:

- *verified mass*, uniform exemplar likelihood for every scheduled bank, including a singleton, with coefficient μ_t ; and
- *known-mode balance*, $\text{KL}(U_{\text{bank}} \| q_{\text{model}})$ for banks with at least two observed modes, with coefficient α_t .

Selection reads only accumulated verified banks and a one-bank compute budget. It never reads exhaustive support, evaluation outcomes, a desired mode count, or reference solutions. Before the first discovery, replay is inactive.

4.4 Projection-free self-reference

The rare-mode, verified-mass, and balance coefficients each warm up for 64 eligible observations and reference only their own warmup mean. Let ρ be normalized open-set entropy, s verified-exemplar surprisal, and h scheduled bank entropy. With EMA decay .9,

$$\beta_t = .10 \frac{\rho_{\text{ref}}}{\rho_{\text{ema}}}, \quad \mu_t = .10 \frac{s_{\text{ema}}}{s_{\text{ref}}}, \quad \alpha_t = .10 \frac{h_{\text{ref}}}{h_{\text{ema}}}.$$

Falling mode entropy increases exploration pressure; rising surprisal on known correct exemplars increases mass pressure; falling bank entropy increases balance pressure. No coefficient is projected. This is a stress-test design: continued collapse remains visible as continued coefficient growth rather than as silent saturation. Singleton banks are omitted from the balance sensor.

4.5 Separated-support escape actuator

All balance mechanisms need an alternative to preserve. A registered extension therefore tests one support-only escape for a singleton bank. It is eligible only after controller warmup, when the run’s own entropy EMA is below its own warmup reference. A domain-specific, validator-preserving transformation may propose at most one new verified key. Proposal rows never enter PPO, and proposal-only support is isolated from the historical count table used by the on-policy rare/new advantages. It may enter replay support; it joins on-policy counts only after the neutral policy independently produces and validates it. A same-plumbing actuator-off arm is the causal comparator.

5 Related Work

Verifier-based reasoning RL. GRPO and Dr.GRPO train reasoning policies from group-relative outcome rewards [11, 3, 10]. Dr.GRPO addresses response-length normalization; our question is orthogonal: what happens to probability *within* the set of equally rewarded correct outcomes?

Diversity collapse and diversity-aware optimization. Recent work documents loss of answer or trajectory diversity under reasoning fine-tuning and RLVR, and intervenes with historical bonuses, within-batch penalties, pass@ K objectives, semantic similarity, divergence changes, set-level objectives, or distribution matching [13, 15, 1, 8, 7, 9, 12]. These approaches optimize complementary objects. MODEBENCH’s distinction is an admission contract: a novelty key exists only if the same generated object executes correctly. xDr.GRPO’s bank is also open-set and policy-generated rather than initialized from a solution catalogue.

Maximum-entropy control. Maximum-entropy RL augments reward with policy entropy and supports adaptive entropy coefficients [16, 6, 4, 5]. xDr.GRPO borrows the principle but changes the random variable: entropy is over validator-positive execution modes, not tokens or a known action set. Its replay and empirical advantages are an online surrogate, not Soft Actor-Critic and not an exact gradient of global sequence entropy.

Executable outputs. Grammar constraints can prevent invalid generations [2]. We instead require execution to bind reward and identity. Python additionally uses a restricted grammar before isolated execution; MathIR uses a finite command language. This reduces false distinctions without claiming that every open-ended reasoning strategy is executable.

6 Experimental Design and Evidence Policy

6.1 Matched training and evaluation

All reported RL runs use Qwen2.5-0.5B-Instruct at a pinned revision, seeds 43–45, group size 16, learning rate 2×10^{-7} , one PPO epoch, $\beta_{\text{KL}} = 0$, rollout temperature 1, top- $p = 1$, maximum response length 192, and at most 12 passes through a fixed training pool. Within a domain, arms share the data, model, optimizer, prompt format, rollout count, training budget, evaluation seed, checkpoint cadence, and recovery rules.

Evaluation uses greedy decoding and four deterministic replicates of $K = 8$ temperature-one samples. Registered ModeBench checkpoints are passes 0, 1, 2, 3, 4, 5, 6, 8, 10, 12. A point appears only when all three seeds in both compared arms have landed it; faster seeds are never carried forward.

6.2 Evidence hierarchy

The registered ModeBench campaign pairs matched Dr.GRPO and xDr.GRPO across four domains, with actuator-off/on comparisons planned across the same benchmark. Infrastructure interruptions left the broad campaign incomplete. We therefore do not report partial AUC, impute missing cells, or call the broad comparison terminal.

Question	Comparison	Status	Allowed claim
Does xDr.GRPO oppose collapse across domains?	4 domains, 2 arms, 3 seeds; latest shared registered checkpoint	exploratory	direction and magnitude at the frozen surface
Does isolated support discovery act causally?	MathIR, same-plumbing actuator off/on, 3 seeds, pass 12	terminal	domain-specific mechanism result

Table 2: **Evidence hierarchy.** Scope is determined by completion and prospective controls, not by whether a result is favorable.

Audit rules. Bank state, controller state, scheduler cursor, optimizer state, data cursor, and request-stream state are checkpointed together. Non-finite values, state mismatch, algorithm exceptions, missing terminal markers, proposal leakage into PPO, or proposal overlap with on-policy counts are fail-closed violations. Placement-only recovery amendments are hash-bound. Two superseded cohorts are excluded by machine-readable objective audits rather than by their outcomes.

7 Results

7.1 Matched Dr.GRPO reaches a single-mode regime

At the frozen shared checkpoints, matched Dr.GRPO has $\text{excess@8} \leq .042$ in every domain (Table 3). Thus, conditional on finding a correct answer, its eight samples almost never contain an additional executed solution. The phenomenon is not simply low accuracy. In Graph coloring, greedy accuracy is .344 both at initialization and pass 10, while excess@8 falls from .190 to .021. In Countdown, accuracy rises substantially but excess breadth remains only .042. Python converges to the same .172 on greedy, mean@8 , pass@8 , and distinct@8 for every seed. MathIR’s distinct@8 and pass@8 are both .364. These are the operational signatures of concentration or failure to escape a singleton correct mode.

7.2 xDr.GRPO improves both search success and executed breadth

Domain	pass	matched Dr.GRPO		xDr.GRPO	
		pass@8	excess@8	pass@8	excess@8
Graph coloring	10	.372	.021	.968	1.376
Countdown	5	.608	.042	.665	1.010
Python factors	6	.172	.000	.740	.216
MathIR action menu	8	.364	.000	.779	.026

Table 3: **Exploratory shared-checkpoint means.** Excess is $\text{distinct@8} - \text{pass@8}$; it counts correct execution modes beyond the first. All entries use three seeds and four fixed sampling replicates.

xDr.GRPO raises pass@8 in every paired seed of every domain at the reported surface. The largest breadth effects are in Graph (+.595 pass@8 , +1.355 excess@8) and Countdown (+.057, +.967). Python improves strongly on average but is seed-unstable: two treatment seeds reach 1.000 pass@8 ,

while the third reaches .219; only one seed produces excess multiplicity at this checkpoint. MathIR more than doubles pass@8, but most of the gain is finding a first correct route rather than multiple routes. The result supports a consistent direction across four validators, not a completed campaign-wide effect estimate.

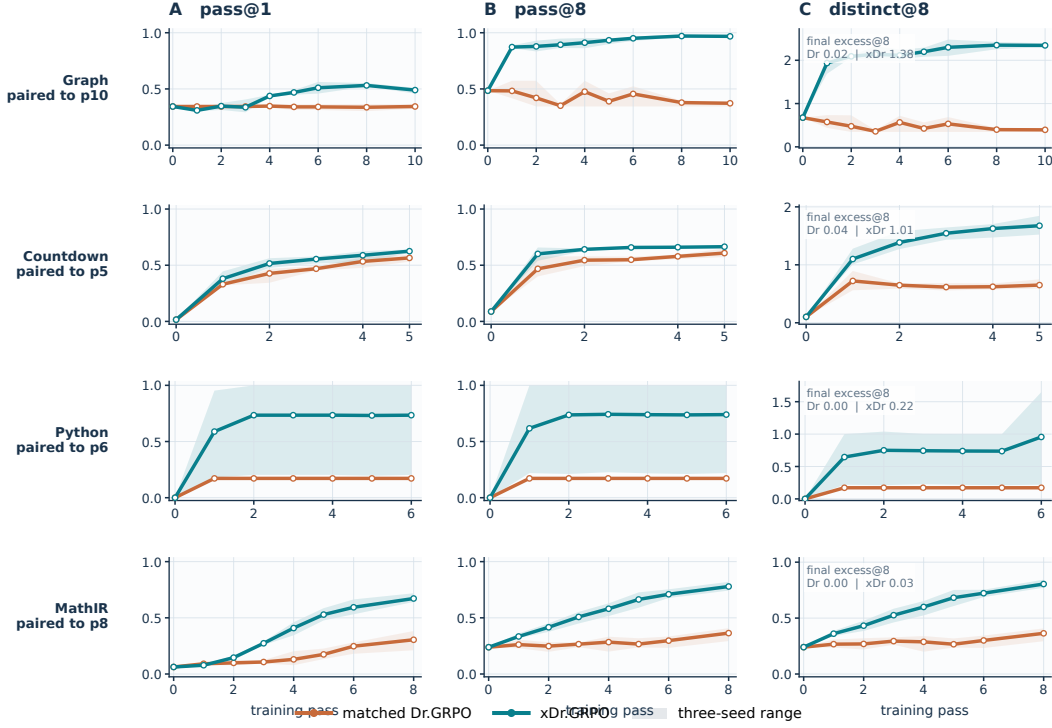


Figure 2: **Mode collapse through training, with metrics separated.** Rows are domains; columns are greedy pass@1, sampled pass@8, and distinct@8, each on its own axis. Lines are three-seed means and ribbons are seed ranges. Final labels report $\text{excess@8} = \text{distinct@8} - \text{pass@8}$. Curves stop at the largest registered checkpoint shared by all seeds of both arms.

7.3 Terminal causal result: separated support discovery

The completed MathIR ablation isolates the singleton actuator from its proposal-capable execution plumbing. Both arms use the same xDr.GRPO objective; only the treatment may place one entropy-gated alternative in separated replay support.

Terminal pass 12 metric	actuator off	separated support	paired mean Δ
greedy pass@1	.7708	.7969	+.0260
mean@8	.7397	.7699	+.0301
pass@8	.8555	.8822	+.0267
distinct@8	.9102	1.0469	+.1367
bank support per prompt	1.3556	1.4821	+.1265

Table 4: **Terminal MathIR causal ablation.** Each mean contains three paired seeds. Greedy, pass@8, and distinct@8 improve in all three pairs; mean@8 improves in two.

All six runs reach the registered 12-pass endpoint and all 49 fixed evaluation points. The treatment records 327 audited interventions. Proposal rows sent to PPO are exactly zero, and proposal-only outcomes remain outside the on-policy count table. The evidence therefore supports the narrow causal statement that separated support discovery improves the xDr.GRPO system on MathIR. Other-domain actuator cells are incomplete and are not claimed.

8 Discussion

Executable identity is central. An outcome-entropy objective is only as meaningful as its equivalence relation: strings reward formatting, while final answers erase alternative procedures. MODEBENCH instead binds identity to the same execution that establishes correctness, and xDr.GRPO optimizes that auditable object.

Discovery, balance, and entropy play different roles. Balancing cannot recover an absent mode, while rewarding every new string invites error variety. The open-set bucket retains unknown support, one-time credit pays for expansion, and replay preserves verified discoveries. Token entropy may aid local exploration, but xDr.GRPO’s empirical bank is specifically an online surrogate for entropy over correct executed outcomes.

The evidence fixes the scope. The shared-checkpoint result motivates the method but is not a completed campaign analysis, and the actuator result is causal only for MathIR. A legitimate next test needs larger models, a fresh complete cohort, and new independently executable multi-mode domains.

9 Limitations and Broader Impact

The broad comparison is incomplete: domains stop at different shared passes, no partial AUC is confirmatory, and three seeds establish paired directions rather than population-level uncertainty. Because the method followed extensive exploratory development, this effect remains hypothesis-generating despite frozen checkpoints and audits.

All four domains are synthetic, the model has 0.5B parameters, and exact mode identity is unusually available. Real proofs and programs can contain meaningful equivalences no canonicalizer resolves. MathIR trades naturalness for exact execution. Outside these environments, unsupported semantic strategy labels remain beyond the scope of our claims.

Python factors admits only a bounded call-free AST, removes builtins, runs in a separate worker on frozen inputs, enforces a timeout, and fails closed. It is not a general code sandbox.

Diversity is not always desirable: valid modes can differ in cost, clarity, robustness, or risk, and a uniform bank can overvalue poor alternatives. Deployments should report correctness, invalidity, breadth, and the mode mapping together, weighting modes by application value when appropriate.

10 Conclusion

Dr.GRPO can improve binary reward while the distribution of correct solutions collapses. We define the failure on verifier-positive execution keys, make it measurable with MODEBENCH, and introduce xDr.GRPO as an online open-set outcome-MaxEnt intervention. The broad surface shows greater search success and breadth, and the terminal MathIR ablation supports separated discovery. Diverse reasoning requires both the metric and the optimizer to act on verified solution modes rather than strings.

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A Why GRPO Collapses and xDr.GRPO Does Not after Coverage

This appendix separates two questions that are easy to conflate: whether a binary-reward group update preserves ratios among already correct modes, and whether xDr.GRPO preserves every correct mode that exists. The first admits an unconditional answer in a finite categorical model. The second necessarily requires a coverage assumption: no method whose bank contains only executed, validator-positive objects can protect a mode that is never sampled or constructed. We therefore prove support preservation first and state full no-collapse as a coverage corollary.

A.1 Categorical reduction and the GRPO mean update

Fix one prompt. Let the finite response categories be $\mathcal{A} = \mathcal{C} \dot{\cup} \mathcal{N}$, where the $m \geq 2$ categories in $\mathcal{C}$ are correct execution modes and the $n \geq 1$ categories in $\mathcal{N}$ are incorrect. Let $p_a = \text{softmax}(z)_a$, $P = \sum_{c \in \mathcal{C}} p_c$, and $q_c = p_c/P$. Thus q is exactly the conditional correct-mode distribution used in the definition of collapse.

Draw a group $A_1, \dots, A_G \stackrel{\text{iid}}{\sim} p$ with $G \geq 2$, write $R_i = \mathbf{1}[A_i \in \mathcal{C}]$, and let $M = \sum_i R_i$. Both GRPO variants can be written, at the on-policy point where the PPO ratio equals one, as

$$\hat{g}_G = \frac{1}{G} \sum_{i=1}^G w(M) \left(R_i - \frac{M}{G} \right) \nabla_z \log p_{A_i}. \quad (1)$$

For Dr.GRPO, $w(M) = 1$ after absorbing the shared positive length scale. For binary-reward GRPO, $w(M)$ is the reciprocal group reward standard deviation on mixed groups. In either case $w(\ell) > 0$ for $\ell \in \{1, \dots, G-1\}$; its value on all-equal groups is irrelevant because the centered advantage is zero. We take explicit reference KL to be zero, as in the experiments.

Lemma A.1 (A group baseline does not remove frequency amplification). *For $0 < P < 1$, the expected group update is*

$$\mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P, \quad c_G(P) = \frac{\mathbb{E}[w(M)M(G-M)]}{G^2 P(1-P)} > 0. \quad (2)$$

For Dr.GRPO, $c_G(P) = (G-1)/G$. Hence reward centering and binary reward standardization change the speed, but not the expected direction, of the binary-reward policy gradient.

Proof. Condition on the complete binary reward vector, whose sum is M . The score identities give

$$\mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 1] = \nabla_z \log P, \quad \mathbb{E}[\nabla_z \log p_{A_i} \mid R_i = 0] = \nabla_z \log(1-P).$$

There are M correct rows with centered reward $(G-M)/G$ and $G-M$ incorrect rows with centered reward $-M/G$. Therefore the conditional expectation of the unaveraged sum in (1) is

$$w(M) \frac{M(G-M)}{G} (\nabla_z \log P - \nabla_z \log(1-P)) = w(M) \frac{M(G-M)}{GP(1-P)} \nabla_z P.$$

The outer factor $1/G$ and expectation over $M \sim \text{Binomial}(G, P)$ yield (2). For $w \equiv 1$, $\mathbb{E}[M(G-M)] = G(G-1)P(1-P)$, proving the Dr.GRPO value. $\square$

A.2 Winner-take-all collapse under GRPO and Dr.GRPO

Consider the infinitesimal expected-update flow $\dot{z} = \mathbb{E}[\hat{g}_G] = c_G(P) \nabla_z P$. This is the clean mean-field limit of fresh on-policy groups and the first-order PPO update. It deliberately removes sampling noise; the result below therefore identifies a structural instability rather than blaming finite batches.

Theorem A.2 (Generic binary-reward GRPO mode collapse). *Start from finite logits with $0 < P(0) < 1$. If the conditional correct-mode distribution $q(0)$ has a unique largest coordinate c^* , then*

$$P(t) \longrightarrow 1, \quad q_{c^*}(t) \longrightarrow 1, \quad q_c(t) \longrightarrow 0 \quad (c \neq c^*).$$

Thus both GRPO and Dr.GRPO converge to a single correct execution mode for generic initial conditions, even though every mode in $\mathcal{C}$ has exactly the same reward.

Proof. For a correct mode c and incorrect category a ,

$$\dot{z}_c = c_G(P)p_c(1 - P), \quad \dot{z}_a = -c_G(P)p_aP.$$

Also $\dot{P} = c_G(P)\|\nabla_z P\|_2^2 > 0$. If P converged to an interior value, continuity and positivity of c_G , together with

$$\|\nabla_z P\|_2^2 \geq P^2(1 - P)^2 \left(\frac{1}{m} + \frac{1}{n} \right),$$

would keep $\dot{P}$ bounded away from zero, a contradiction. Hence $P \rightarrow 1$.

Define the effective optimization time

$$\tau(t) = \int_0^t c_G(P(s))P(s)(1 - P(s)) ds.$$

If $\tau(\infty) < \infty$, every correct and incorrect logit would have finite total variation, because the absolute velocity of each is at most $\dot{\tau}$. All logits would then converge to finite values, contradicting $P \rightarrow 1$. Therefore $\tau(t) \rightarrow \infty$.

On the correct simplex, changing from t to τ gives the replicator flow

$$\frac{dq_c}{d\tau} = q_c \left(q_c - \sum_{d \in \mathcal{C}} q_d^2 \right), \quad \frac{d}{d\tau} \log \frac{q_c}{q_d} = q_c - q_d. \quad (3)$$

The initially unique maximum remains the unique maximum, and every one of its log-ratios increases. The Lyapunov quantity $\sum_c q_c^2$ is nondecreasing, since its derivative is twice the variance of the coordinate values under q . Every limit point of (3) is therefore uniform on its positive support. Because $q_{c^*} \geq 1/m$ and its ratio to every other coordinate stays strictly greater than one, no multi-coordinate uniform limit is possible. The only limit point is the vertex $q_{c^*} = 1$. $\square$

Corollary A.3 (Ties are nongeneric). *If exactly $k < m$ correct modes tie for the largest initial probability, symmetry preserves the tie and the flow converges to the uniform distribution on those k modes while eliminating the rest. Only the exactly uniform initialization over all m correct modes avoids collapse. Any arbitrarily small perturbation that creates a unique maximum reduces to Theorem A.2.*

Finite groups make this instability operationally harsher: a mode absent from a group has no score carrier, all-correct groups have zero task advantage, and sampling noise almost surely breaks exact ties. These facts are not needed for the theorem.

A.3 Why the xDr.GRPO terms reverse the instability

The open-set actor signal has a direct maximum-entropy interpretation before clipping and count approximation.

Lemma A.4 (Centered surprisal is the execution-mode entropy gradient). *Let $C \sim q$ on a known correct support and $\mathcal{H}(q) = -\sum_c q_c \log q_c$. Then*

$$\mathbb{E}_{C \sim q} [(-\log q_C - \mathcal{H}(q)) \nabla_z \log q_C] = \nabla_z \mathcal{H}(q).$$

Thus, when the detached predictor equals the current correct-mode distribution, the centered rare-mode advantage is an unbiased entropy-gradient estimator.

Proof. The baseline vanishes by the score identity. For the remaining term,

$$\sum_c q_c (-\log q_c) \nabla_z \log q_c = -\sum_c (\log q_c) \nabla_z q_c = \nabla_z \mathcal{H}(q),$$

because $\sum_c \nabla_z q_c = 0$ supplies the missing constant in the derivative of $-q_c \log q_c$. $\square$

The implemented pseudocount, unseen bucket, and surprisal clip make this a robust surrogate rather than an exact identity. The one-time new-mode term adds total credit ν independent of the number of duplicates once a new key appears. Neither signal can assign gradient to a never-observed zero-carrier

mode; that is why verified replay and the coverage condition below matter. The next result needs only the balance term; it does not count any additional restoring pressure from rare-mode or verified-mass replay. We index time by mean updates acting on this prompt, absorbing fixed round-robin frequency and replay scaling into the effective coefficient $a(t)$; the categorical abstraction excludes cross-prompt parameter interference.

Let a verified bank contain $m \geq 2$ modes, let $u_c = 1/m$, and let q be the model distribution normalized over those bank modes. In the categorical projection, descending the implemented balance loss $L(q) = \text{KL}(u||q)$ contributes $a(t)(u - q)$ to the mode logits. Binary GRPO contributes the amplifying term $b(t)q$, where $b(t) = c_G(P)P(1 - P)$ in the mean flow. Terms shared by every correct-mode logit are denoted $d(t)\mathbf{1}$ and do not change q .

Theorem A.5 (Verified balance makes the uniform bank the unique attractor). *Suppose that after some time T the bank is fixed and scheduled infinitely often, and its mean conditional logit dynamics are*

$$\dot{z} = b(t)q + a(t)(u - q) + d(t)\mathbf{1}. \quad (4)$$

If $a(t) > b(t)$ eventually and $\int_T^\infty (a(t) - b(t)) dt = \infty$, then

$$\text{KL}(u||q(t)) \longrightarrow 0, \quad q(t) \longrightarrow u.$$

Consequently every bank mode retains asymptotic probability $1/m$ and conditional mode collapse is impossible on that bank.

Proof. Let $\lambda(t) = a(t) - b(t)$. Since $u = (1/m)\mathbf{1}$, Equation (4) is, up to a common-logit shift,

$$\dot{z} = -\lambda(t)(q - u) + \tilde{d}(t)\mathbf{1}.$$

For $L(z) = \text{KL}(u||\text{softmax}(z))$, $\nabla_z L = q - u$ and $(q - u)^\top \mathbf{1} = 0$. Hence

$$\dot{L} = -\lambda(t)\|q - u\|_2^2 \leq 0. \quad (5)$$

Use replay time $s = \int_T^t \lambda(r)dr$, which diverges by assumption. Equation (5) is ordinary gradient flow for L modulo the irrelevant common-logit direction. Its sublevel sets are compact in the open simplex because $\text{KL}(u||q) \rightarrow \infty$ at the boundary, and its only stationary distribution is $q = u$. Therefore $q(t) \rightarrow u$ and $L(t) \rightarrow 0$. $\square$

Corollary A.6 (xDr.GRPO no-collapse guarantee after verified coverage). *Assume the categorical mean-flow model above and:*

1. *every correct execution mode enters and remains in the verified bank by a finite time—so the protected support fits within replay capacity;*
2. *deterministic global replay visits that bank infinitely often;*
3. *the balance controller is active with reference entropy $h_{\text{ref}} > 0$ and tracks bank entropy through $a(t) = a_0 h_{\text{ref}} / h_{\text{ema}}(t)$; and*
4. *correct mass approaches one, so the GRPO amplification $b(t) = c_G(P)P(1 - P)$ approaches zero.*

Then $q(t) \rightarrow u$ on the full correct support. In particular,

$$\min_{c \in \mathcal{C}} q_c(t) \longrightarrow \frac{1}{m}, \quad \mathcal{H}(q(t)) \longrightarrow \log m,$$

so mode collapse cannot emerge asymptotically.

Proof. Because $0 < h_{\text{ema}}(t) \leq \log m$ on eligible replay observations, $a(t) \geq a_0 h_{\text{ref}} / \log m =: a_{\min} > 0$. Since $b(t) \rightarrow 0$, eventually $a(t) - b(t) \geq a_{\min}/2$, whose integral over infinitely many replay visits diverges. Theorem A.5 applies. Moreover, if entropy attempted to approach zero before that regime, $h_{\text{ema}} \rightarrow 0$ would make $a(t) \rightarrow \infty$ while $b(t)$ remains bounded, so the boundary cannot be an attracting limit. $\square$

Remark A.7 (Exact scope of the guarantee). If the verified bank covers only $\mathcal{B} \subsetneq \mathcal{C}$ —including when the true support exceeds the implementation’s capacity 16—the theorem guarantees non-collapse and asymptotic balance on $\mathcal{B}$, not discovery of $\mathcal{C} \setminus \mathcal{B}$. The open-set unseen bucket, first-discovery credit, and separated-support actuator are mechanisms for satisfying coverage; they are not a catalogue oracle. Likewise, the proof concerns realizable categorical mean dynamics with recurrent replay. It does not claim global convergence for an arbitrary nonconvex language model. This distinction is what makes the guarantee falsifiable rather than rhetorical.

B External Python Verification Contract

A Python response is parsed in expression mode and must be a lambda with one argument named n . Accepted nodes are integer constants, n , bounded arithmetic, comparisons, Boolean operators, unary operators, and conditional expressions. Calls, imports, attributes, comprehensions, containers, subscripts, assignment, loops, and every other identifier are rejected before execution. Candidate length and AST size are bounded.

The parent sends restricted source and trusted JSON task data to a persistent external interpreter over JSON lines. The worker reparses, removes builtins, evaluates the lambda, and invokes it on all frozen inputs under a per-request timer. It returns only validity, the integer vector, and its key. Timeout, broken pipe, malformed output, non-integer values, or divisor failure returns no key. The parent can restart a failed worker.

For inputs $[6, 10, 15]$,

```
lambda n: 2 if n % 2 == 0 else 3
```

returns $(2, 2, 3)$, while

```
lambda n: 3 if n % 3 == 0 else 5
```

returns $(3, 5, 3)$. Both verify and receive different behavior keys.

C Dataset and Identity Audits

	Python factors	MathIR action menu
Training rows	384	384
Evaluation rows	128	128
Exact modes per prompt	16–3,600	5
Training-row SHA-256	bcfa9acc...37cf7	bfc98362...5d3835
Evaluation-row SHA-256	be0f621c...94d183	df5783dd...328b96
Gold catalogue supplied to policy	no	no
Training-bank initialization	empty	empty

Table 5: Deterministic materialization identities. For every Python row, two independently executed functions must verify with different keys. Every MathIR certified route is executed by the same interpreter used for model outputs.

D Algorithmic Details

Each optimizer update follows this order:

1. sample one on-policy group and freeze the prompt’s pre-group bank;
2. execute every response, intersect validation with positive reward, and assign keys only to agreed validator-positive rows;
3. compute the centered Dr.GRPO task advantage;
4. compute rare-mode advantages under the detached open-set predictor;
5. split one discovery credit across duplicates of each first-seen key;
6. optionally generate one eligible separated-support proposal, revalidate it, and keep it outside PPO and on-policy counts;

7. commit neutral discoveries to the count bank after group scoring;
8. choose one accumulated bank by checkpointed global round robin;
9. add verified-mass replay and, for multi-key banks, balance replay; and
10. update the policy, controller EMAs, bank, and scheduler cursor, then checkpoint all states together.

With one scheduled bank and replay capacity 16, the implementation scales the replay contribution as

$$\frac{15}{16} \frac{1}{16} (\mu_t L_{\text{mass}} + \alpha_t L_{\text{balance}}).$$

The historical prompt-local bank-entropy channel is disabled (`bank_alpha=0`); the open-set advantage is the sole outcome-entropy channel in the actor advantage. Direct token entropy is absent from the final objective.

E Per-Seed Shared-Checkpoint Results

Domain (pass)	Method	seed	pass@8	mean@8	distinct@8	pass@1
Graph (10)	Dr.GRPO	43	.344	.344	.344	.344
		44	.388	.327	.409	.323
		45	.385	.361	.427	.365
	xDr.GRPO	43	.971	.396	2.336	.500
		44	.964	.397	2.302	.458
		45	.969	.406	2.393	.510
Countdown (5)	Dr.GRPO	43	.648	.594	.750	.594
		44	.572	.537	.588	.531
		45	.604	.565	.613	.570
	xDr.GRPO	43	.664	.559	1.846	.609
		44	.672	.576	1.521	.625
		45	.658	.577	1.656	.641
Python (6)	Dr.GRPO	43	.172	.172	.172	.172
		44	.172	.172	.172	.172
		45	.172	.172	.172	.172
	xDr.GRPO	43	1.000	.995	1.000	1.000
		44	1.000	.993	1.648	1.000
		45	.219	.188	.219	.203
MathIR (8)	Dr.GRPO	43	.408	.384	.408	.383
		44	.293	.205	.293	.211
		45	.391	.338	.391	.320
	xDr.GRPO	43	.773	.632	.799	.656
		44	.820	.721	.844	.719
		45	.742	.580	.771	.641

Table 6: Seed-level values underlying Table 3. Each block is at the largest registered checkpoint shared by all seeds of both arms. Values are rounded to three decimals.

F Terminal MathIR Causal Rows

arm	seed	greedy	mean@8	pass@8	distinct@8	bank support
actuator off	43	.7500	.7207	.8379	.8418	1.2924
	44	.7578	.6980	.8438	.9355	1.4167
	45	.8047	.8005	.8848	.9531	1.3577
separated support	43	.7578	.7246	.8477	.9824	1.4646
	44	.8125	.7900	.9023	1.1035	1.4922
	45	.8203	.7949	.8965	1.0547	1.4896

Table 7: The three paired terminal MathIR rows. All runs end at step 4,608, corresponding to 12 passes.

G Design Failures that Constrain the Final Method

Observed failure	Scientific consequence	Retained decision
Sequence entropy increased length and EOS avoidance	token uncertainty can grow without solution breadth	optimize validator-bound outcomes
Candidate-local normalization favored short responses	a nominally uniform target introduced length gradients	keep mode pressure as a separate advantage
Detached candidate fitting optimized a sampled target	reweighting candidates is not policy MaxEnt	state the empirical outcome surrogate explicitly
Group-centered surprise vanished under concentration	a controller cannot amplify a zero-dispersion carrier	center under the predictive bank distribution
Novel wrong answers received diversity pressure	error variety is not useful breadth	admit only validator-positive outcomes
Fixed or projected pressure saturated	the coefficient bound became the intervention	self-reference controllers with no projection
Prompt-local replay left discoveries idle	late alternatives could not act promptly	schedule one verified bank per update
A shared proposal/count bank leaked off-policy support	proposals changed the next on-policy advantage	separate replay support from objective counts

Table 8: Exploratory development determines the final design but is not pooled into its treatment effect.

H Clean Cohort Progress and Causal Provenance

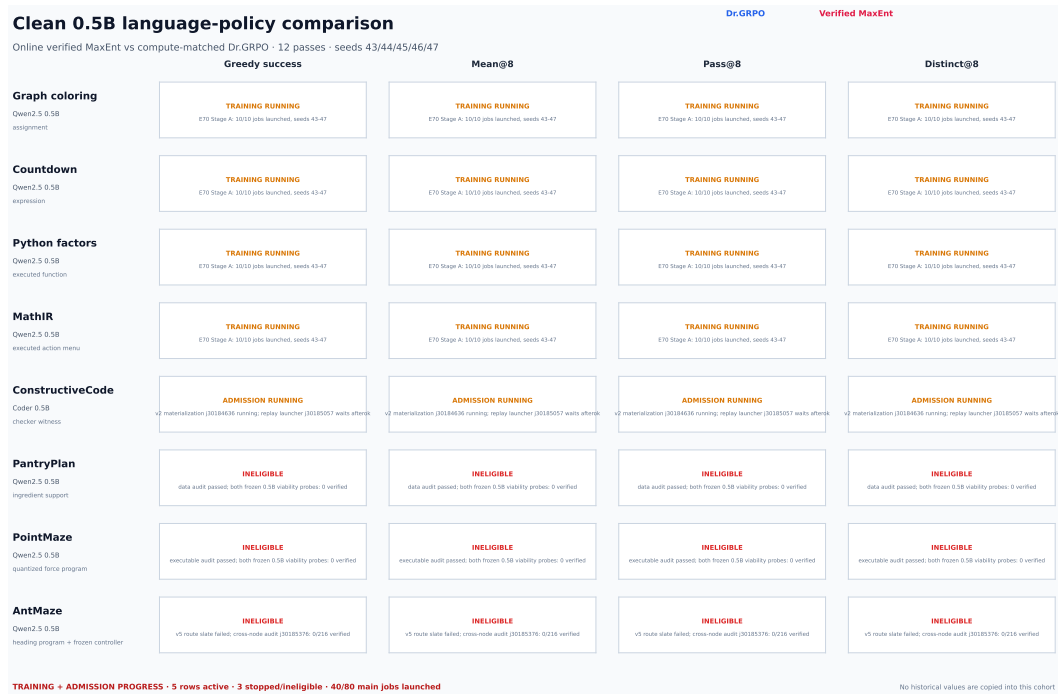


Figure 3: Registered MATH-free eight-environment 0.5B comparison surface. At this progress snapshot, E70 Stage A has launched the two frozen arms and five seeds on the four established ModeBench domains; ConstructiveCode v2 admission is running, while PantryPlan, PointMaze, and AntMaze remain explicitly stopped at their prospective gates. Empty metric panels contain no copied historical result.

I Reproducibility and Claim Boundaries

Figure 1 is generated by `ops/plot_paper_collapse_toy.py` directly from the retained seed-43 Qwen2.5-0.5B-Instruct sample logs. Its prompt-selection rule, displayed counts, source paths, and source hashes are frozen in `var/artifacts/paper_graph_collapse_toy.json`. The aggregate figures are generated by `ops/plot_paper_modecollapse.py` from `var/artifacts/e65_five_domain_confirmation_results_latest.json`. Table 4 is reproduced from `var/artifacts/e66_e68_mathir_terminal_recovery_summary.json`. The clean eight-row progress surface is rendered by `ops/exp_scaling/plot_clean_05b_eight_environment_cohort.py` from `var/artifacts/clean_05b_eight_environment_progress.json`, with hashes in `var/artifacts/clean_05b_eight_environment_figure_provenance.json`. The broad historical comparison protocol is `paper/preregistration/e61r1_e58_vs_grpo_05b_12pass.md`. The clean eight-row surface and its running four-row Stage-A block are bound by `docs/clean_05b_maxent_vs_drgrpo_eight_environment_plan.md` and `paper/preregistration/e70_clean_verified_maxent_vs_compute_matched_drgrpo_stage_a_05b.md`; the objective is `paper/preregistration/e58_global_verified_replay_canonical_05b.md`; the same-plumbing control and separated-support treatment are `paper/preregistration/e66_same_plumbing_actuator_ablation_05b.md` and `paper/preregistration/e68_separated_support_actuator_ablation_05b.md`.

The environments are implemented in `src/oat_drgrpo/canonical_actions.py`, `src/oat_drgrpo/python_modebench.py`, and `src/oat_drgrpo/mathir.py`. The external Python process boundary is in `src/oat_drgrpo/python_modebench_process.py` and its worker in `src/oat_drgrpo/python_modebench_worker.py`. The open-set signal is in `src/oat_drgrpo/semantic_shannon.py`, replay in `src/oat_drgrpo/canonical_replay.py`, the bank in `src/oat_drgrpo/online_canonical_bank.py`, and the separated actuator in `src/oat_drgrpo/counterfactual_proposals.py`.

The manuscript separates four evidence classes:

1. exploratory design evidence, used to motivate components but never pooled into the final method comparison;
2. the incomplete registered ModeBench campaign, reported only at paired shared preregistered checkpoints;
3. the terminal six-run MathIR causal ablation; and
4. invalidated or failed successor cohorts, retained in the repository but excluded from every headline estimate.